FACULTY
OF MATHEMATICS
AND PHYSICS
Charles University

Visual Task Modeling and Automated Planning for Humanoid Robots in Education of Individuals with Autism Spectrum Disorder

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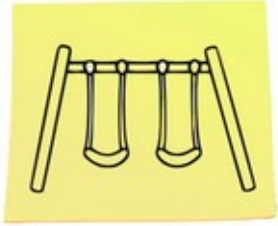
Robot-assisted education of people with ASD

- persons with Autism Spectrum Disorder (ASD):
  - respond positively to interactions with robots
  - may benefit from robot predictability and consistency
  - potential to enhance learning outcomes via robot-assisted education

TEACCH® Autism Program

- persons with ASD:
  - are visual learners
  - prefer visually cued instructions

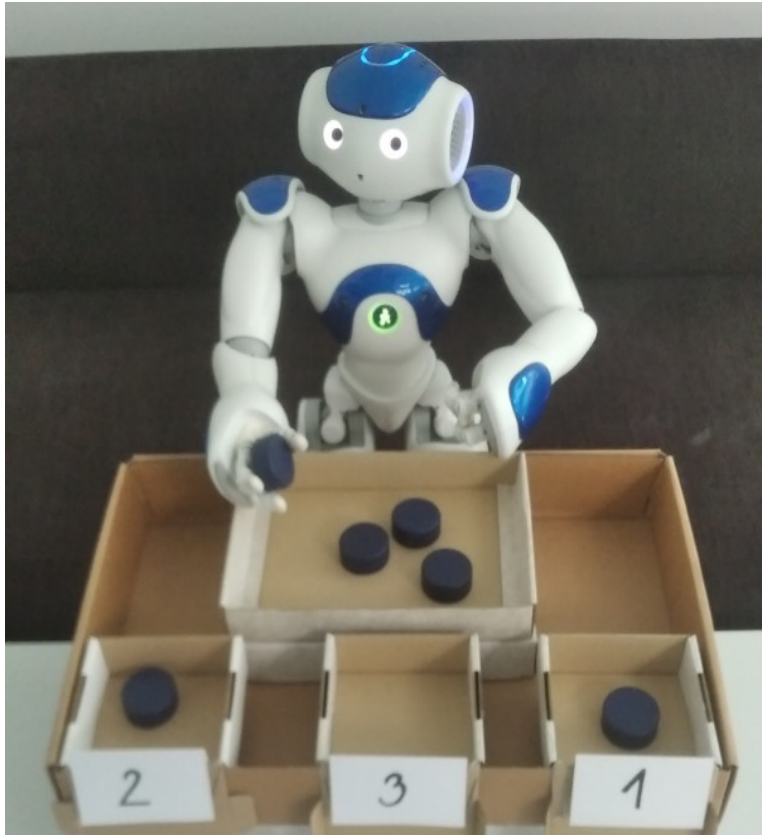


- visually structured tasks
- visual schedules

Treatment and Education of Autistic and related Communication-handicapped Children

What is a Shoebox Task?

- structured educational task

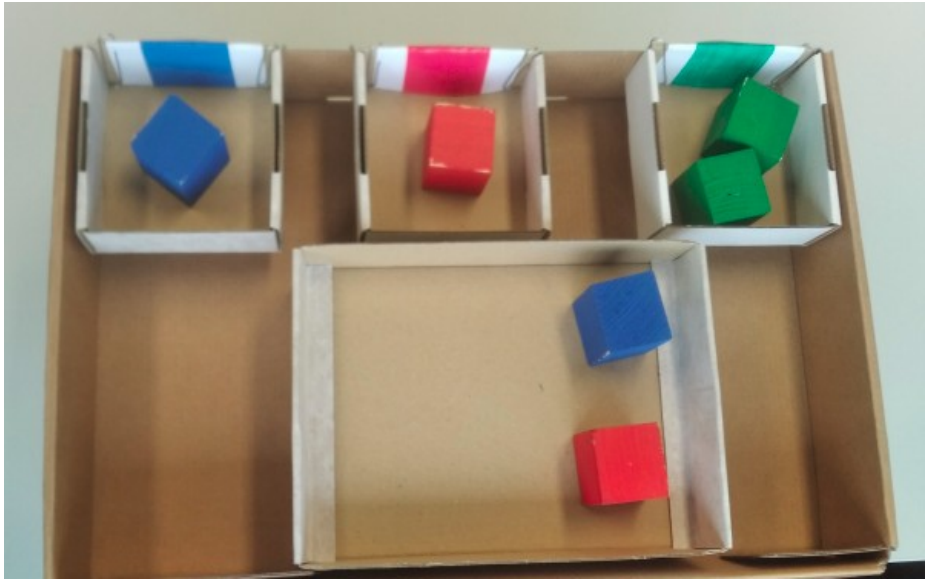


What is a Shoebox-Task?

- the goal can be deduced from **the visual structure**
- no further instructions needed

ShoeboxTasks®

DIY Shoebox-Task



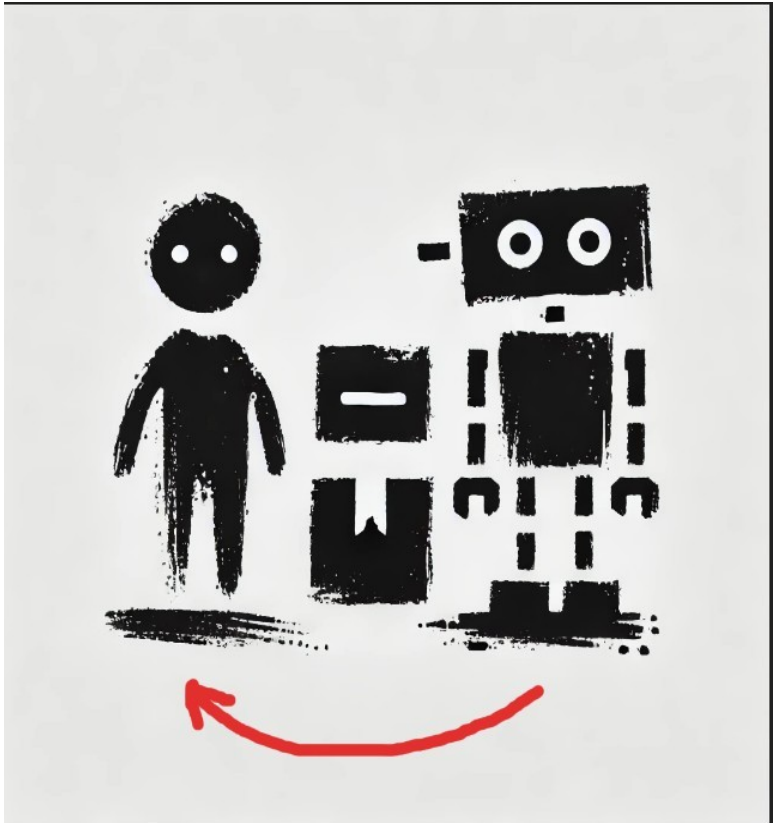
How to solve a Shoebox-Task?

- the goal can be deduced from the **visual structure**

- no further instructions needed

- perfect for automatic **vision-based** analysis

1. Robot as a Teacher



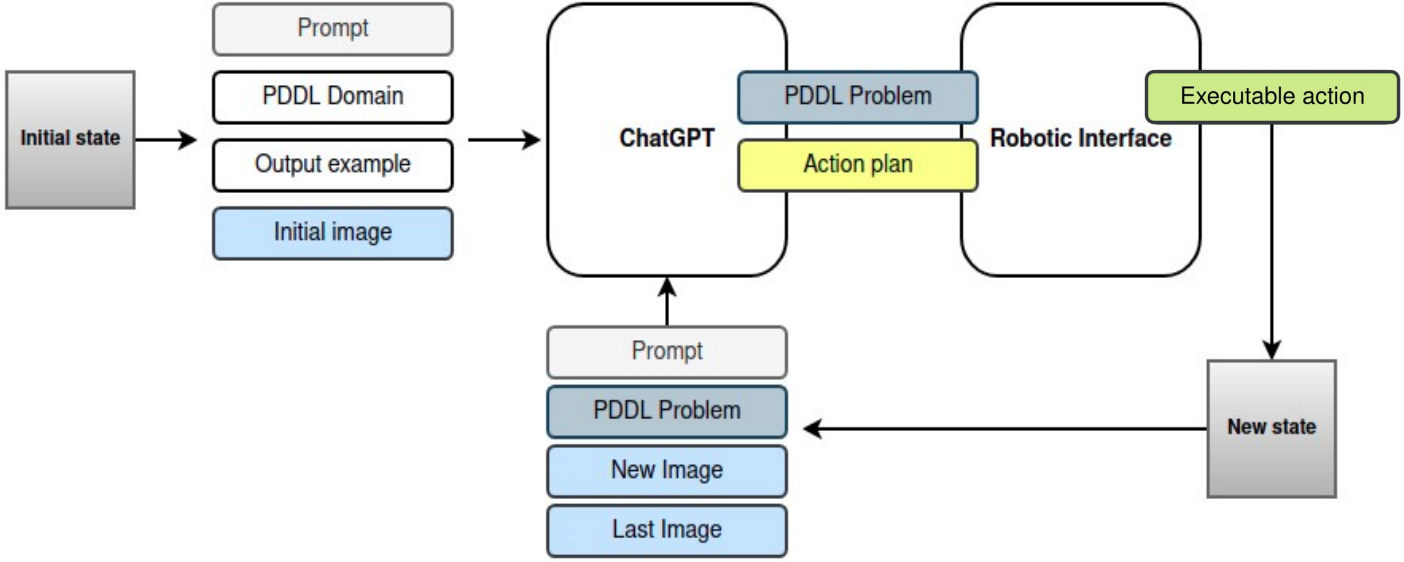
1. Robot as a Teacher

- What does the robot need to understand the task?
  - Analysis and planning
  - **Task solver**
- How can the robot guide the student?
  - Speech, gestures, demonstration, …
  - **Robotic interface**

Task Solver: Automated Analysis

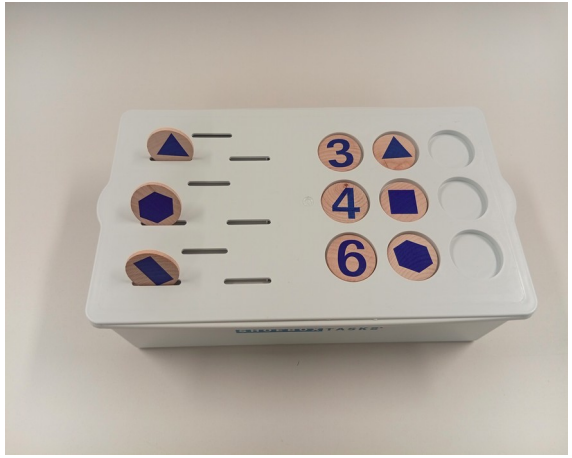
- Initial state:
  - PDDL representation + action plan
- Next states:
  - Recognize and validate performed actions
  - Update the plan accordingly

Task Solver: Overview



Initial state:

PDDL representation + action plan

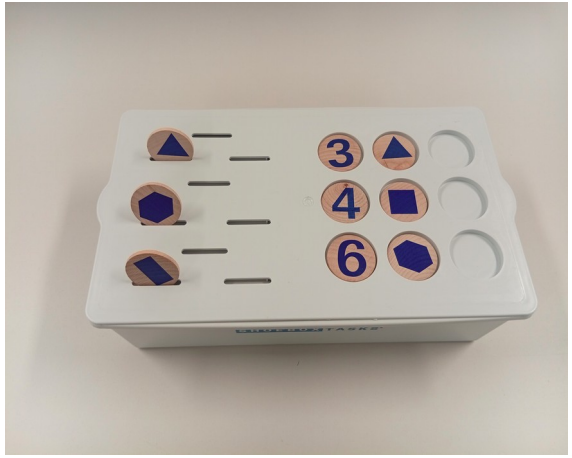



```
(define (problem shape_matching_problem)
  (:domain shoebox-matching)
  (:objects
  )
  (:init
  )
  (:goal
  )
)
; PLAN

; Summary:
; ….
```

1. Robot as a Teacher

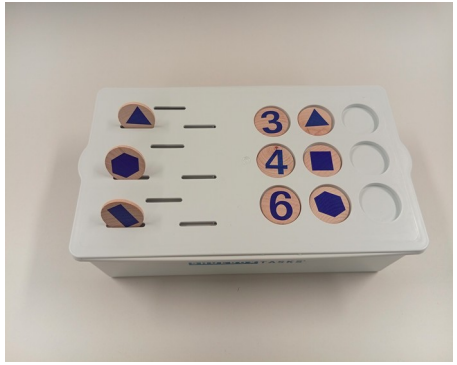
Initial state:

PDDL representation + **action plan**




```
; move_to_target triangle_item target_3

; move_to_target rhombus_item target_4

; move_to_target hexagon_item target_6


; Summary:

; Total number of objects: 6

; Object names: triangle_item, hexagon_item, rhombus_item,
target_3, target_4, target_6

; No uncertainties or partially visible items

; Task goal: Move each shaped item from the init_container to
its corresponding target based on matching shapes and
positions.
```

Next states:




```
; move_to_target triangle_item target_3

; move_to_target rhombus_item target_4

; move_to_target hexagon_item target_6
```

```
; move_to_target rhombus_item target_4
```

Robotic Interface

Action

Executable action

move_to_target ?i - item ?t -target

```
pick_and_place(item_coord, target_coord)
```

Robotic Interface

Action

- **move_to_target item_red target_red**

Executable action

```
pick_and_place(item_red_coord, target_red_coord)
```

- `pick(i_coord)`
- `move_hand_above(i_coord)`
- `open_gripper()`
- `move_hand_to(i_coord)`
- `close_gripper()`
- `place(t_coord)`
- `...`

Robotic Interface

Executable action

Experiments

colors-3

colors-3-diy

colors-4

shapes-angles

numbers

shapes

Experiments

TABLE I
SHOEBOX TASK PERFORMANCE METRICS: CHATGPT-4O

Shoebox Task	Time (s)	Semantic Errors	Inaccurate Names	Syntactic Errors	Model Refusals
colors-3	10.9	0	0	0	1
colors-3-diy	8.9	0	0	0	0
colors-4	12.0	0	0	0	0
shapes-angles	11.8	1	10	0	0
numbers	11.2	0	0	0	0
shapes	8.5	0	1	0	0

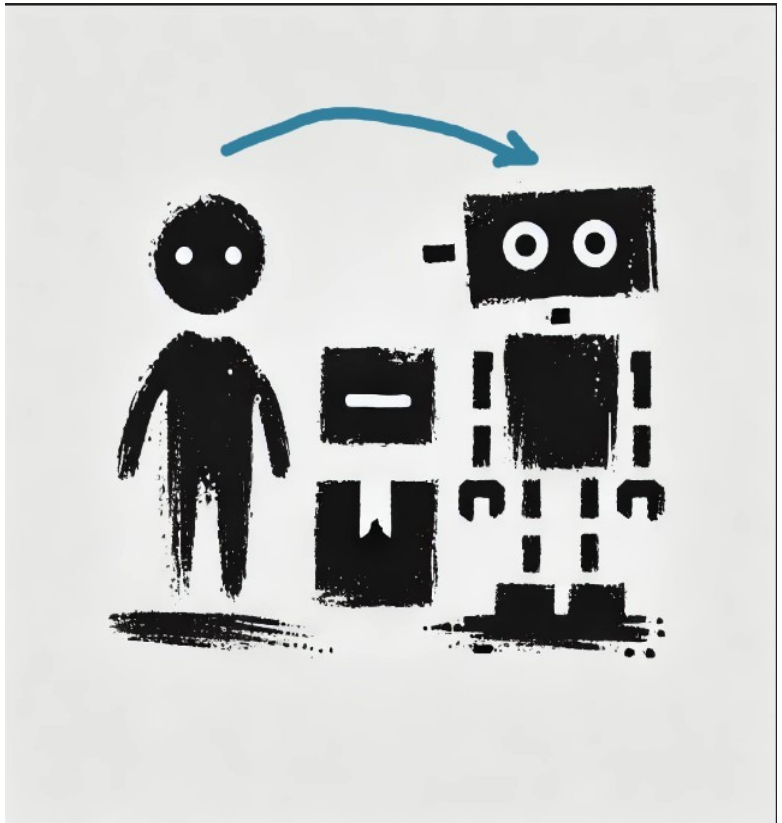
[a]All times are averages over ten trials, rounded to two decimal places.

TABLE II
SHOEBOX TASK PERFORMANCE METRICS: CHATGPT-5

Shoebox Task	Time (s)	Semantic Errors	Inaccurate Names	Syntactic Errors	Model Refusals
colors-3	75.8	0	0	0	0
colors-3-diy	40.7	0	0	0	0
colors-4	49.5	0	0	0	0
shapes-angles	86.4	1	1	0	0
numbers	42.9	0	0	0	0
shapes	61.7	0	0	0	0

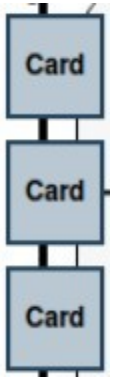
[a]All times are averages over ten trials, rounded to two decimal places.

2. Student Guiding the Robot



Visual programming interface

Action

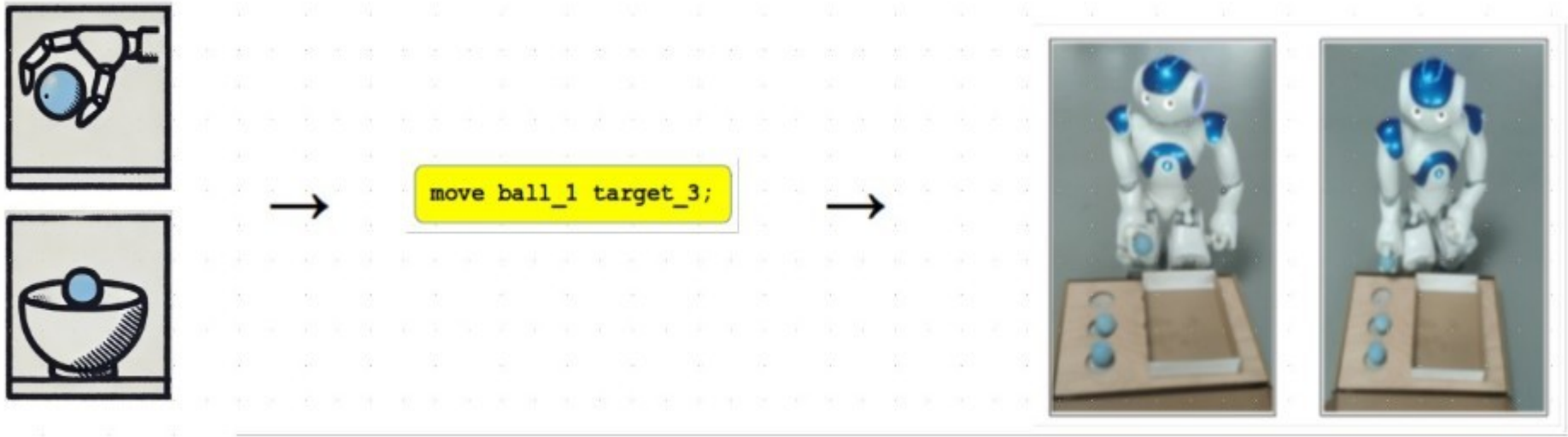


Executable action

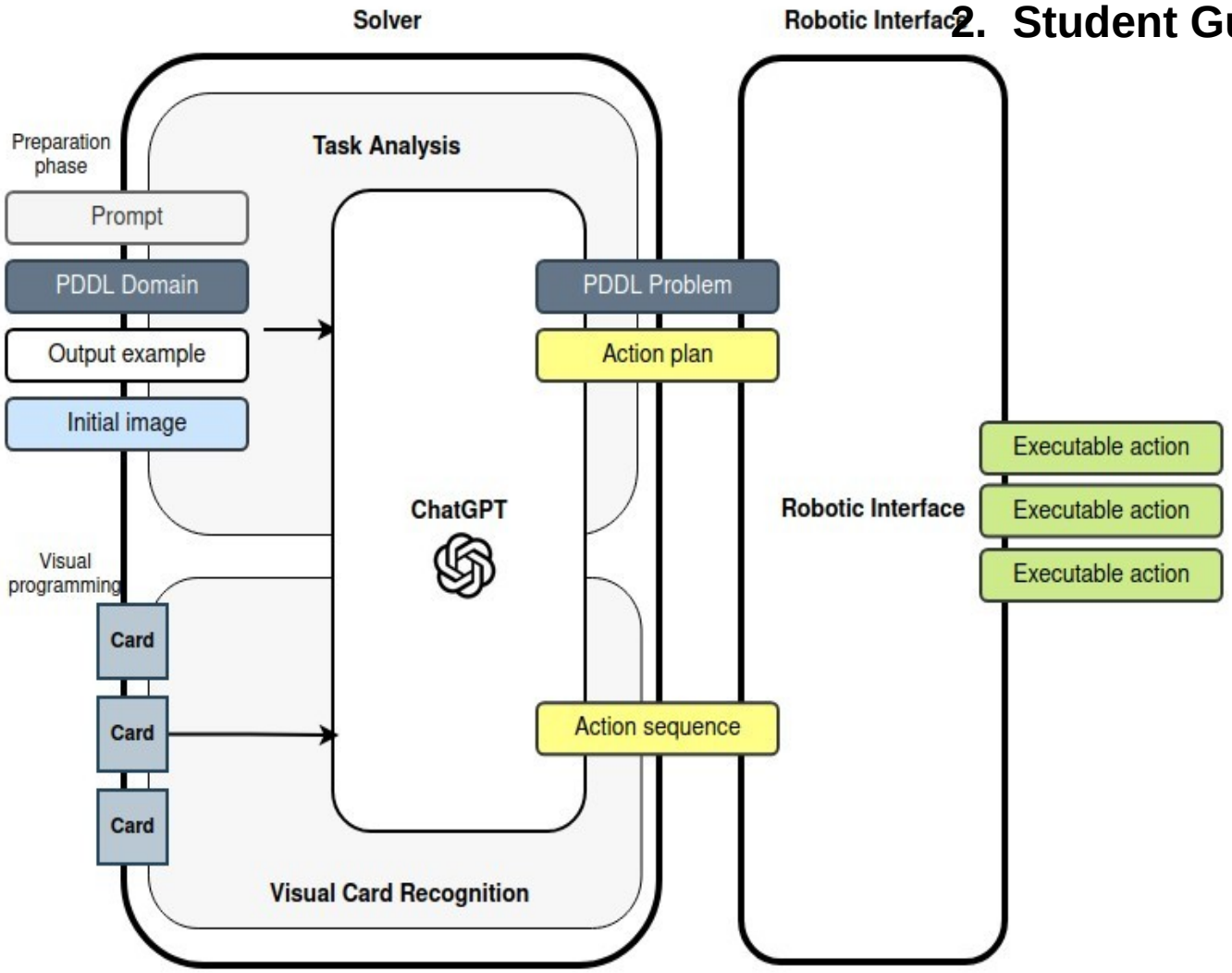
```
pick_and_place(item_red_coord, target_red_coord)
```

- `pick(i_coord)`
- `move_hand_above(i_coord)`
- `open_gripper()`
- `move_hand_to(i_coord)`
- `close_gripper()`
- `place(t_coord)`
- `...`

Visual programming interface



2. Student Guiding the Robot



Collaboration

- Ron Larsen
  - https://www.shoeboxtasks.com/

- Dave Catlin
  - Educational Robot: Roamer

SHOEBOX TASKS
HANDBOOK
C.O.D.

Contact

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- https://github.com/LadaKa/shoebox-task-solver-results

Slide
THANK
YOU